ICSE Class X Chemistry

Chapter-wise Question Bank: Chapters 8–9

Shekhar Kumar Jha · ICSE 2027 · 24 October 2026

How to use this: Attempt every question cold on paper. Mark only after finishing a section. Log each wrong or blank item as a gap, rewrite it, and retry it after 24 hours. Source tags: **T23–T26** = board paper of that year; **S27 / S26 / S24** = specimen paper of that year. Items without a tag come from the syllabus and are board-style.

Chapter 8: Study of Compounds

A. Hydrogen Chloride: theory

8.1 Laboratory preparation of HCl from sodium chloride: reactants, equations for both stages (below and above 200 °C), labelled apparatus, drying agent, method of collection and test for the gas. [T24, T23]

8.2 Why can HCl gas not be dried over quicklime? Why can it not be collected over water? [T23, S26, S24]

8.3 Fountain experiment with HCl: describe the set-up, observations and the two properties it demonstrates. Why is the fountain red when litmus is used?

8.4 Preparation of hydrochloric acid by dissolving HCl gas in water: draw the inverted funnel arrangement and explain how back-suction is avoided.

8.5 Write equations for HCl reacting with: ammonia, a metal (Mg or Zn), a metal oxide, a hydroxide, a carbonate, silver nitrate solution and lead nitrate solution.

8.6 Give the composition and uses of aqua regia. Give the tests for HCl gas and for hydrochloric acid.

A. Hydrogen Chloride: objective and application

8.7 In the laboratory preparation, HCl gas is dried by passing it through: (a) dilute nitric acid (b) concentrated sulphuric acid (c) dilute sulphuric acid (d) acidified water [T23]

8.8 The property exhibited by concentrated sulphuric acid when used to prepare HCl from potassium chloride: (a) dehydrating (b) drying (c) oxidising (d) non-volatile acid property [T23]

8.9 Non-metals like chlorine and iodine, on reaction with hydrogen, give: (a) acid (b) base (c) salt (d) metal [S27]

8.10 HCl gas is prepared by the action of conc. sulphuric acid on sodium chloride. (a) Give the balanced equation. (b) State the method of collection. (c) Which property of sulphuric acid makes it a suitable reagent? [T24]

B. Ammonia: theory

8.11 Laboratory preparation of ammonia from ammonium chloride: reactants, equation, labelled apparatus, drying agent (and why conc. H_2SO_4, $CaCl_2$ and P_2O_5 cannot be used), method of collection and test.

8.12 Other preparations: from nitrides (Mg_3N_2, AlN) with warm water and from ammonium salts with alkalis. Give equations. Why is the nitride method seldom used? [T23, T25]

8.13 Haber's process: balanced equation, temperature, pressure, catalyst and promoter, and how ammonia is separated. [T26, S24]

8.14 Fountain experiment with ammonia: set-up, observation and the two properties demonstrated. [T26]

8.15 Write equations for ammonia reacting with: HCl, hot CuO, PbO_2, chlorine (excess Cl_2 and excess NH_3), oxygen (burning), and the catalytic oxidation of ammonia.

8.16 Aqueous ammonia: write equations with H_2SO_4, HNO_3, HCl and the solutions of $FeCl_3$, $FeSO_4$, $Pb(NO_3)_2$, $Zn(NO_3)_2$ and $CuSO_4$.

8.17 Give the uses of ammonia (fertilisers, explosives, nitric acid, refrigerant and its non-ozone-depleting alternatives, cleansing agents) and the tests for ammonia.

B. Ammonia: objective questions

8.18 On passing ammonia over heated copper oxide, a reddish-brown residue is left. The property of ammonia demonstrated: (a) basic (b) oxidising (c) reducing (d) acidic [T24]

8.19 Ammonia is collected by downward displacement of air since it is: (a) very slightly soluble in water (b) heavier than air (c) lighter than air (d) insoluble in water [T24]

8.20 Compound P is heated with NaOH solution and a red litmus paper at the mouth of the test tube turns blue. P could be: zinc sulphate, copper sulphate, ferrous sulphate or ammonium sulphate. [S24]

8.21 Ammonia is passed through quicklime and collected in a jar. Red and blue litmus papers are placed in the jar. W: red turns blue, blue remains blue. X: red remains red, blue remains blue. Y: red remains red, blue turns red. Z: red turns blue, blue turns red. Which is correct? [S24]

8.22 Match the reaction with its catalyst: (a) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (b) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (c) $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (d) $C_2H_4 + H_2 \rightarrow C_2H_6$ (e) $CuSO_4{\cdot}5H_2O \rightarrow CuSO_4 + 5H_2O$. Catalysts: vanadium pentoxide, nickel, iron, conc. sulphuric acid, platinum. [T25]

8.23 Identify P in CuO + P → Cu + water + nitrogen, and R in ammonia + R → nitrogen + water. [S27]

8.24 Name the alkaline gas which produces dense white fumes when reacted with HCl gas. [S24]

8.25 Rewrite by adding the correct word: (a) Sulphuric acid acts as a dehydrating agent. (b) Ammonia reacts with chlorine to give ammonium chloride and nitrogen. [T24]

8.26 Name the gas produced when: (a) copper is treated with hot conc. nitric acid (b) ammonia is burnt in an atmosphere of oxygen (c) ferrous sulphide reacts with dil. HCl. [T25]

B. Ammonia: application

8.27 Laboratory preparation of ammonia from magnesium nitride: (a) balanced equation (b) why is the method seldom used? (c) how do you identify the gas? [T23]

8.28 Ammonia is prepared from aluminium nitride in the laboratory. (a) Name the substance X added through the thistle funnel. (b) Write the balanced equation. [T25]

8.29 Ammonia is passed over heated copper(II) oxide in a combustion tube. (a) Name the gas evolved. (b) What is the colour of the residue left? [T25]

8.30 Test tube A has sodium chloride and B has ammonium chloride. NaOH solution is added to both and warmed. (a) In which tube is a gas evolved? (b) Name the gas. (c) What is the colour of the flame when this gas burns in oxygen? (d) Write the equation for the reaction. [S27]

8.31 Give one significant observation when excess chlorine gas reacts with ammonia. [S26, S27]

8.32 Give a reason: when ammonia gas is passed over black copper oxide in a combustion tube, a reddish-brown substance is left behind. [S26]

8.33 Give one observation: (a) burning of ammonia in excess of oxygen (b) dry ammonia gas is passed over heated PbO. [T23]

8.34 Write the balanced equation, with conditions, for the conversion of ammonia to nitric oxide using oxygen and platinum catalyst. [T23]

8.35 Complete and balance: $NH_4Cl + Ca(OH)_2 \rightarrow$ [S26, T23]

8.36 Ranjana passes ammonia gas over heated copper oxide to prove ammonia is a reducing agent. (a) What will she observe? (b) Write the balanced equation. [S24]

8.37 For the industrial manufacture of ammonia: (a) name the process (b) name the catalyst (c) how is ammonia separated from the unreacted gases to give liquid ammonia? (d) which two properties of ammonia does the fountain experiment show? [T26]

8.38 For Haber's process give (a) the temperature of the reaction (b) the catalyst used (c) the balanced equation. [S24]

C. Nitric Acid: theory

8.39 Laboratory preparation of nitric acid from KNO_3 or $NaNO_3$: reactants, equation, conditions, labelled apparatus, precautions (all-glass apparatus, temperature) and collection.

8.40 Ostwald's process: write the three stages with equations and conditions.

8.41 Nitric acid as an oxidising agent: give equations for its reaction with copper (conc. and dilute), carbon and sulphur.

8.42 Acidic properties of nitric acid: give equations with magnesium, oxides, hydroxides, carbonates, hydrogen carbonates and sulphites.

8.43 Give the test for nitric acid. Why is nitric acid stored in amber bottles and why does it turn yellow on standing?

C. Nitric Acid: objective and application

8.44 The salt formed when concentrated sulphuric acid reacts with KNO_3 above 200 °C: (a) K_2SO_4 (b) K_2SO_3 (c) $KHSO_4$ (d) $KHSO_3$ [T23]

8.45 The catalyst used in Ostwald's process: finely divided iron, graphite, vanadium pentoxide or platinum? [S24]

8.46 Which is NOT true of nitric acid? (a) It is a strong reducing agent (b) It is a strong oxidising agent (c) It is unstable to heat (d) It liberates SO_2 when treated with potassium sulphite [S24]

8.47 Seema adds copper turnings to concentrated acid P and a reddish-brown gas evolves. (a) Name acid P. (b) Write the balanced equation. [T25]

8.48 Carbon reacts with an acid to form carbon dioxide, water and nitrogen dioxide. (a) Name the acid. (b) Write the balanced equation. [T26]

8.49 Anil stored nitric acid in a transparent container exposed to sunlight. Brown fumes appeared and the acid turned yellowish. (a) What was his error? (b) Why were brown fumes seen? (c) How can the yellow tinge be removed? [T26]

8.50 In $NaNO_3 + H_2SO_4 \rightarrow NaHSO_4 + HNO_3$: (a) give one reason for keeping the temperature below 200 °C (b) why is concentrated H_2SO_4 used? [T26]

8.51 Nitric acid reacts with compound Q to give $Ca(NO_3)_2$, water and carbon dioxide. Identify Q and write the balanced equation. [S24, S26]

8.52 Write balanced equations for: (a) copper reacting with cold dilute nitric acid (b) laboratory preparation of nitric acid using sodium nitrate. [S27]

8.53 Why does nitric acid normally not liberate hydrogen when it reacts with active metals? [S26]

8.54 Write the equation for the oxidation of carbon with conc. nitric acid. [S24, S26]

8.55 Write the equation for the oxidation of sulphur with hot concentrated nitric acid. [S27, T25]

D. Sulphuric Acid: theory

8.56 Contact process: raw materials, the three stages with equations, catalyst, temperature and pressure. Why is SO_3 not dissolved directly in water? What is oleum?

8.57 Dilute sulphuric acid as an acid: write equations with a metal, metal oxide, hydroxide, carbonate, hydrogen carbonate, sulphite and sulphide. Why is it called dibasic?

8.58 Concentrated sulphuric acid as (a) an oxidising agent (carbon, sulphur) (b) a dehydrating agent (sugar, copper sulphate crystals, ethanol) (c) a non-volatile acid (NaCl, KCl, $NaNO_3$, KNO_3). Give equations.

8.59 Give the tests for sulphuric acid and for the sulphate ion.

D. Sulphuric Acid: objective and application

8.60 A non-metal that reacts with concentrated sulphuric acid to form two gases which turn lime water milky: sulphur, carbon, oxygen or nitrogen? [T26]

8.61 In $S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$ sulphuric acid acts as: (a) non-volatile acid (b) dibasic acid (c) oxidising agent (d) reducing agent [T25]. Which is the oxidising agent in the same equation: sulphur, sulphuric acid, sulphur dioxide or water? [T24]

8.62 Assertion (A): In the Contact process SO_3 is not directly dissolved in water to obtain sulphuric acid. Reason (R): Dense fog or misty droplets of sulphuric acid form, which are difficult to condense. [T25]

8.63 Glucose reacts with concentrated sulphuric acid to give sugar charcoal. The reaction is: oxidation, combustion, dehydration or combination? [S24]

8.64 In the Contact process, what is used in step R to obtain sulphuric acid from sulphur trioxide? (a) vanadium pentoxide (b) water only (c) water followed by conc. H_2SO_4 (d) conc. H_2SO_4 followed by water [S27]

8.65 An industrial process of an acid: name of the process (a), reactants $SO_2 + O_2$, catalyst (b), final product (c). Identify (a), (b), (c). [T24]

8.66 What property of sulphuric acid is shown in: (a) preparation of HCl from NaCl (b) conc. H_2SO_4 reacting with copper to produce SO_2? [S24]

8.67 State the property of sulphuric acid in: (a) sulphur with conc. sulphuric acid (b) conversion of ferrous sulphide to H_2S using sulphuric acid (c) ethanol with conc. sulphuric acid. [S26]

8.68 Differentiate between cane sugar and hydrated copper sulphate using conc. H_2SO_4. [T23]

8.69 A chemist collected the gas formed after catalytic oxidation of SO_2. (a) Name the gas. (b) Which acid forms when it dissolves directly in water? (c) Why should it not be dissolved directly in water? [T26]

8.70 Write the equation for conc. sulphuric acid added to hydrated copper sulphate. [S26]

8.71 A blue crystalline salt B is treated with a few drops of a dense oily hygroscopic liquid. A colourless neutral vapour is released and the salt turns white. (a) Identify B. (b) Which property of the liquid is shown? (c) Write the balanced equation. [S27]

8.72 Write the balanced equation for the reaction of concentrated sulphuric acid with carbon. [T25]

Chapter 9: Organic Chemistry

A. Theory and structures

9.1 Why is carbon unique? Define tetravalency and catenation. Give examples of single, double and triple bonds, straight chain, branched chain and cyclic compounds (benzene structure).

9.2 Draw structural formulae of alkanes, alkenes and alkynes up to 5 carbon atoms. Draw all isomers of butane and pentane.

9.3 Define isomerism. Explain chain and position isomerism with examples. [T25]

9.4 Define homologous series and state its characteristics. Give the general formulae of alkanes, alkenes and alkynes. [T23]

9.5 State the IUPAC rules (longest chain, smallest number for the functional group). Name compounds with halide, double bond, triple bond, alcohol, aldehyde, ketone, carboxylic acid, ether and ester groups, and give trivial names such as acetic acid, chloroform and acetylene.

9.6 Methane and ethane: preparation from sodium ethanoate and sodium propanoate with soda lime, and from iodomethane and bromoethane; complete and incomplete combustion; chlorination (substitution) with equations.

9.7 Ethene: preparation by dehydrohalogenation and by dehydration; addition reactions with H_2, Cl_2, Br_2. Ethyne: preparation from calcium carbide and from 1,2-dibromoethane; addition reactions. Give a test for unsaturation.

9.8 Give the uses of methane, ethane, ethene and ethyne.

9.9 Ethanol: preparation by hydrolysis of an alkyl halide; physical properties; chemical properties (combustion, reaction with sodium, esterification, dehydration); denatured, methylated and spurious spirit; uses. [T25]

9.10 Acetic acid: structure, physical properties (vinegar odour, glacial acetic acid), chemical properties (litmus, alkalis, alcohol/esterification) and uses.

B. Objective questions

9.11 Which formula represents a saturated hydrocarbon? (a) C_4H_8 (b) C_5H_{12} (c) C_4H_6 (d) C_5H_{10} [T25]

9.12 Which set has compounds from the same homologous series? (a) CH_4, C_2H_6, C_3H_8 (b) CH_4, C_2H_6, C_3H_6 (c) C_3H_4, C_3H_6, C_3H_8 (d) C_2H_4, C_3H_6, C_4H_{10} [T25]

9.13 Which equation shows the reaction of ethane with chlorine in UV light? (a) $C_2H_6 + Cl_2 \rightarrow C_2H_6Cl_2$ (b) $C_2H_6 + Cl_2 \rightarrow C_2H_4Cl_2 + H_2$ (c) $C_2H_6 + Cl_2 \rightarrow C_2H_5Cl + HCl$ (d) $C_2H_6 + Cl_2 \rightarrow 2CH_3Cl$ [T26]

9.14 An unsaturated hydrocarbon with three carbon atoms and six hydrogen atoms is: propyne, propane, propene or propanol? [T26]

9.15 Unsaturated hydrocarbons undergo: addition, substitution, oxidation or redox reactions? [T24]

9.16 Reaction of an alcohol with a carboxylic acid in the presence of conc. H_2SO_4 is: halogenation, esterification, hydrogenation or dehydrohalogenation? [T24]

9.17 Conversion of ethanol to ethene by conc. sulphuric acid involves: dehydration, dehydrogenation, dehydrohalogenation or hydrolysis? [T24]

9.18 An example of a cyclic organic compound: propene, pentene, butene or benzene? [T23]

9.19 The hydrocarbon formed when sodium propanoate and soda lime are heated: methane, ethane, ethene or propane? [T23]

9.20 General formula of hydrocarbons with single covalent bonds: C_nH_{2n+2}, C_nH_{2n}, C_nH_{2n-2} or C_nH_{2n-6}? [T23]

9.21 The functional group in methanol: >C=O, –OH, –CHO or –COOH? [S24]

9.22 Methanol and ethanol belong to the same homologous series. This means: (a) only C and H atoms (b) same number of C atoms (c) same functional group (d) same relative molecular mass [S26]

9.23 Organic compound(s) with molecular formula C_3H_8O: 1. propanol 2. propanal 3. propanoic acid. (a) only 1 (b) only 2 (c) 2 and 3 (d) 1 and 2 [S27]

9.24 The general formula of an alkyl group: C_nH_{2n+1} or C_nH_{2n-1}? [S27] Alkanes undergo addition or substitution reactions? [S24]

9.25 From ethene, ethanoic acid, ethanol and methanal choose: (a) the compound with no double bond (b) the one which in pure form turns into an ice-like solid on cooling (c) the one used for artificial ripening of fruits. [T24]

9.26 Arrange C_2H_2, C_3H_6, CH_4, C_2H_4 in increasing order of molecular weight. [T23]

9.27 Conversion of ethene to ethane is an example of hydration or hydrogenation? [T23]

9.28 True or False: Organic compounds having different molecular formula but the same structural formula are called isomers. [T26]

9.29 Name: (a) organic compounds with the same molecular formula but different structural formulae (b) the group of compounds whose successive members differ by CH_2 (c) undistilled alcohol containing a large amount of methanol. [T23, T25]

C. Equations and application

9.30 Complete and balance: $C_2H_4Br_2$ + KOH (alcoholic) → [T26]

9.31 Complete and balance: (a) CH_3COONa + NaOH (CaO, heat) → (b) CH_3COOH + Mg → [T25]

9.32 Complete and balance: (a) C_2H_5COONa + NaOH (CaO, heat) → (b) $C_2H_4Br_2$ + alcoholic KOH → [T24]

9.33 Give balanced equations with conditions for: (a) ethene from ethanol (b) ethyne from calcium carbide (c) monochloromethane from methane. [T23]

9.34 An unsaturated hydrocarbon with two carbon atoms reacts with two moles of hydrogen in the presence of a catalyst to give a saturated hydrocarbon. (a) Name the type of reaction. (b) Name a suitable catalyst. (c) Write the balanced equation. [T26]

9.35 Identify: (a) a compound that reacts with water to give acetylene (b) the gas evolved when sodium propanoate is heated with soda lime. [T26]

9.36 Give balanced equations for: (a) dehydration of ethanol by conc. H_2SO_4 at 170 °C (b) laboratory preparation of ethyne from calcium carbide (c) conversion of acetic acid to ethyl acetate. [S24]

9.37 Give reasons: (a) ethene undergoes addition reactions (b) hydrocarbons can be used as fuels. [S24]

9.38 Write the equation when bromine gas is passed over ethene in the presence of carbon tetrachloride. [S26]

9.39 Give the equation for the laboratory preparation of ethanol using chloroethane and aqueous sodium hydroxide. [S26]

9.40 Write the balanced equation for chlorination of chloroform in the presence of diffused sunlight. [S27]

9.41 Give balanced equations for: (a) action of warm water on calcium carbide (b) laboratory preparation of ethane using sodium propionate and soda lime. [S27]

9.42 A mixture of ethanol, acetic acid and conc. H_2SO_4 is heated in a round-bottom flask. (a) Name the type of reaction. (b) State the role of sulphuric acid. (c) State one observation. [S27, S26]

9.43 Differentiate between unsaturated and saturated hydrocarbons on the basis of the type of bond present. [T24]

9.44 Name the group obtained by removing one hydrogen atom from the parent alkane. [T24]

9.45 Draw the structural diagram of each: 2-methyl propene; butanal; pent-2-yne; the isomer of n-butane [T25, T26]. 3-methyl pentane; propyne; methanal [T24]. 2-chlorobutane; but-2-yne [T23]. 1-propanal; 1,2-dichloroethane; but-2-ene [S24]. 1,1,1-trichloromethane; pentan-2-ol [S26]. diethyl ether; isopentane; propanone [S27].

9.46 Give the IUPAC name of: $CH_3–CH(CH_3)–CH_3$; $CH_3CH_2CH(OH)CH_3$; CH_3CH_2COOH; CH_3CH_2CHO; CH_3COCH_3; $CH_3CH_2OCH_2CH_3$; CH_3COOCH_3.

Answer Key

Objective and numerical items only. Write theory items from your own notes, then check them against your textbook.

Chapter 8: Hydrogen Chloride

8.7 (b) concentrated sulphuric acid

8.8 (d) non-volatile acid property

8.9 (a) acid

8.10 (a) $NaCl + H_2SO_4 \rightarrow NaHSO_4 + HCl$ (below 200 °C) (b) upward displacement of air (downward delivery), since HCl is heavier than air and very soluble in water (c) non-volatile nature

Chapter 8: Ammonia

8.18 (c) reducing

8.19 (c) lighter than air

8.20 ammonium sulphate

8.21 W

8.22 (a) iron (b) platinum (c) vanadium pentoxide (d) nickel (e) conc. sulphuric acid

8.23 P = ammonia; R = oxygen

8.24 ammonia

8.25 (a) Concentrated sulphuric acid acts as a dehydrating agent. (b) Excess ammonia reacts with chlorine to give ammonium chloride and nitrogen.

8.26 (a) NO_2 (b) nitrogen (c) H_2S

8.27 (a) $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$ (b) magnesium nitride is costly and not easily available (c) turns moist red litmus blue; forms dense white fumes with HCl

8.28 (a) water (warm) (b) $AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$

8.29 (a) nitrogen (b) reddish-brown (copper)

8.30 (a) B (b) ammonia (c) greenish-yellow (d) $NH_4Cl + NaOH \rightarrow NaCl + NH_3 + H_2O$

8.31 a yellow oily explosive liquid (NCl_3) forms

8.32 ammonia is a reducing agent: it reduces CuO to reddish-brown copper

8.33 (a) it burns with a greenish-yellow flame (b) yellow PbO turns to grey lead; nitrogen is evolved and water condenses

8.34 $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ (platinum catalyst, about 800 °C)

8.35 $2NH_4Cl + Ca(OH)_2 \rightarrow CaCl_2 + 2H_2O + 2NH_3$

8.36 (a) black CuO turns reddish-brown; colourless gas (nitrogen) evolves; water droplets form (b) $3CuO + 2NH_3 \rightarrow 3Cu + 3H_2O + N_2$

8.37 (a) Haber's process (b) finely divided iron (c) ammonia liquefies on cooling; unreacted N_2 and H_2 are recycled (d) very high solubility in water and alkaline nature

8.38 (a) about 450–500 °C (b) finely divided iron (molybdenum as promoter) (c) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ (about 200 atm)

Chapter 8: Nitric Acid

8.44 (a) K_2SO_4

8.45 platinum

8.46 (a) it is a strong reducing agent

8.47 (a) nitric acid (b) $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

8.48 (a) nitric acid (b) $C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$

8.49 (a) stored in a transparent container in sunlight; it should be kept in a dark (amber) bottle in a cool place (b) light decomposes it: $4HNO_3 \rightarrow 4NO_2 + 2H_2O + O_2$ (c) warm gently or bubble air through to drive off dissolved NO_2

8.50 (a) above 200 °C the nitric acid decomposes (and $NaHSO_4$ reacts further with $NaNO_3$) (b) conc. H_2SO_4 is non-volatile with a high boiling point, so it displaces volatile HNO_3

8.51 Q is calcium carbonate: $CaCO_3 + 2HNO_3 \rightarrow Ca(NO_3)_2 + H_2O + CO_2$

8.52 (a) $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 4H_2O + 2NO$ (b) $NaNO_3 + \text{conc. } H_2SO_4 \rightarrow NaHSO_4 + HNO_3$ (below 200 °C)

8.53 it is a strong oxidising agent and oxidises the hydrogen formed to water

8.54 $C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$

8.55 $S + 6HNO_3 \rightarrow H_2SO_4 + 6NO_2 + 2H_2O$

Chapter 8: Sulphuric Acid

8.60 carbon ($C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$)

8.61 (c) oxidising agent; sulphuric acid

8.62 (a) both true, R explains A

8.63 dehydration

8.64 (d) conc. H_2SO_4 followed by water

8.65 (a) Contact process (b) vanadium pentoxide, V_2O_5 (c) sulphuric acid

8.66 (a) non-volatile acid (b) oxidising agent

8.67 (a) oxidising (b) acidic (c) dehydrating

8.68 cane sugar chars to black carbon; blue hydrated copper sulphate turns white

8.69 (a) SO_3 (b) sulphuric acid (c) it forms a dense fog of acid droplets that is hard to condense, and the reaction is very violent

8.70 $CuSO_4{\cdot}5H_2O \rightarrow CuSO_4 + 5H_2O$ (conc. H_2SO_4 removes the water)

8.71 (a) $CuSO_4{\cdot}5H_2O$ (b) dehydrating property (c) $CuSO_4{\cdot}5H_2O \rightarrow CuSO_4 + 5H_2O$ (conc. H_2SO_4)

8.72 $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$

Chapter 9: Organic Chemistry

9.11 C_5H_{12}

9.12 (a)

9.13 (c)

9.14 propene

9.15 addition

9.16 esterification

9.17 dehydration

9.18 benzene

9.19 ethane

9.20 C_nH_{2n+2}

9.21 –OH

9.22 (c)

9.23 (a) only 1

9.24 C_nH_{2n+1}; substitution

9.25 (a) ethanol (b) ethanoic acid (c) ethene

9.26 CH_4 (16) < C_2H_2 (26) < C_2H_4 (28) < C_3H_6 (42)

9.27 hydrogenation

9.28 False: isomers have the same molecular formula but different structural formulae

9.29 (a) isomers (b) homologous series (c) spurious alcohol

9.30 $C_2H_4Br_2 + 2KOH\ (alc.) \rightarrow C_2H_2 + 2KBr + 2H_2O$

9.31 (a) $CH_3COONa + NaOH \rightarrow CH_4 + Na_2CO_3$ (b) $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$

9.32 (a) $C_2H_5COONa + NaOH \rightarrow C_2H_6 + Na_2CO_3$ (b) $C_2H_4Br_2 + 2KOH \rightarrow C_2H_2 + 2KBr + 2H_2O$

9.33 (a) $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (conc. H_2SO_4, 170 °C) (b) $CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$ (c) $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ (diffused sunlight)

9.34 (a) addition (b) nickel (c) $C_2H_2 + 2H_2 \rightarrow C_2H_6$

9.35 (a) calcium carbide (b) ethane

9.36 (a) $C_2H_5OH \rightarrow C_2H_4 + H_2O$ (b) $CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$ (c) $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$ (conc. H_2SO_4)

9.37 (a) ethene is unsaturated (C=C), so it adds atoms across the double bond (b) hydrocarbons burn in air releasing a large amount of heat

9.38 $C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$

9.39 $C_2H_5Cl + NaOH\ (aq) \rightarrow C_2H_5OH + NaCl$

9.40 $CHCl_3 + Cl_2 \rightarrow CCl_4 + HCl$

9.41 (a) $CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$ (b) $C_2H_5COONa + NaOH \rightarrow C_2H_6 + Na_2CO_3$ (soda lime, heat)

9.42 (a) esterification (b) dehydrating agent (catalyst) (c) fruity smell

9.43 unsaturated hydrocarbons have double or triple bonds; saturated hydrocarbons have only single bonds

9.44 alkyl group

9.46 2-methylpropane; butan-2-ol; propanoic acid; propanal; propanone; ethoxyethane; methyl ethanoate

Next step: This completes the Class X Chemistry syllabus. Now go to full-length timed papers from T23-T26 and the 2027 specimen.